SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –** Pseudocode Design	**Assessment:**	**Assessment Tool 1–** Pseudocode Design
Weightage:	**30% of CIA**	**Total Marks:**	**100**
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
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Section / Batch:	2024- CSE(AI)	**Date:**	25-08-26

Code of Conduct

I __S.Sirivally / 192472371__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sirivally.S___

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. **Design pseudocode to implement the K-Means clustering algorithm: initialize centroids, assign observations to the nearest centroid, update centroids, and repeat until convergence.**

Problem Statement

Develop a procedure to implement the K-Means clustering algorithm for grouping similar observations into a predefined number of clusters. The algorithm should begin by selecting initial centroids for the required number of clusters. Each observation is assigned to the nearest centroid based on a distance measure. The centroids are then recalculated using the observations assigned to each cluster. This assignment and updating process should continue until the centroids become stable or the algorithm reaches the maximum number of iterations.

Algorithm / Procedure

1. Choose the number of clusters, **K**.
2. Initialize K centroids randomly.
3. Calculate the distance between every observation and each centroid.
4. Assign every observation to its nearest centroid.
5. Recalculate each centroid using the mean of its assigned observations.
6. Repeat the assignment and centroid-update steps until the centroids no longer change.
7. Display the final clusters and their centroids.

Pseudocode

```
START
Input dataset X
Choose number of clusters K
Initialize K centroids randomly
REPEAT
   FOR each observation x in X
      Calculate distance from x to every centroid
      Assign x to the nearest centroid
   END FOR
   FOR each cluster k
      Calculate the mean of all observations
      assigned to cluster k
```

```
        Update centroid k
    END FOR
UNTIL centroids do not change
       OR maximum iterations are reached
Display final clusters
Display final centroids
STOP
```

Output

Number of clusters: 3

Final Centroids:
Cluster 1 → Centroid = (...)
Cluster 2 → Centroid = (...)
Cluster 3 → Centroid = (...)

Cluster assignments:
Observation 1 → Cluster 2
Observation 2 → Cluster 1
Observation 3 → Cluster 3
...

2. **Design pseudocode to group customers into three clusters based on annual income and spending score using K-Means clustering**

Problem Statement

Develop a K-Means clustering procedure to segment customers based on their annual income and spending score. The objective is to divide customers into three groups with similar purchasing characteristics. Annual income and spending score are used as the two clustering features. K-Means assigns customers to the nearest cluster based on these features. The resulting clusters can be used to understand different customer segments and support targeted marketing decisions.

Algorithm / Procedure

1. Load the customer dataset.
2. Select Annual Income and Spending Score as features.
3. Set K = 3.
4. Initialize three centroids.
5. Assign each customer to the nearest centroid.
6. Update the centroids using the mean values of each cluster.
7. Repeat until convergence.
8. Display the three customer segments.

Pseudocode

```
START
Load customer dataset
Select:
   Annual_Income
   Spending_Score
Set K = 3
Initialize 3 centroids
REPEAT
   FOR each customer
      Calculate distance to each centroid
      Assign customer to nearest centroid
   END FOR
```

FOR each cluster
    Calculate mean Annual_Income
    Calculate mean Spending_Score
    Update centroid
END FOR
UNTIL centroids become stable
Display cluster assignment for each customer
Display final cluster centroids
STOP

Output

Customer Segmentation

Cluster 1:
Annual Income → Low/Medium
Spending Score → High

Cluster 2:
Annual Income → High
Spending Score → High

Cluster 3:
Annual Income → Medium/High
Spending Score → Low
The exact cluster characteristics depend on the dataset.

3. **Design pseudocode to determine the suitable number of clusters using the Elbow Method and then apply K-Means with the selected number of clusters.**

Problem Statement

Develop a procedure to determine a suitable number of clusters for a dataset using the Elbow Method. Different values of K are tested by applying K-Means clustering and calculating the within-cluster sum of squares (WCSS). The WCSS generally decreases as the number of clusters increases. The suitable value of K is selected at the point where adding additional clusters produces only a small improvement. K-Means is then applied using the selected number of clusters.

Algorithm / Procedure

1. Load and prepare the dataset.
2. Select a suitable range of K values, such as 1 to 10.
3. Apply K-Means for every K value.
4. Calculate WCSS for each K.
5. Plot K against WCSS.
6. Identify the elbow point.
7. Select the K value at the elbow.
8. Apply K-Means using the selected K.
9. Display the final clusters.

Pseudocode

```
START
Load dataset X
FOR K = 1 to 10
   Apply K-Means with K clusters
   Calculate WCSS
   Store WCSS value
END FOR
Plot K versus WCSS
Identify the elbow point
Select K at the elbow
```

Apply K-Means using selected K
Display final clusters
STOP

Output

WCSS values:

K = 1 → 2500
K = 2 → 1400
K = 3 → 800
K = 4 → 600
K = 5 → 520
...

Elbow point = K = 3

Selected number of clusters = 3

The main output is an Elbow Plot, where the sharp decrease in WCSS begins to level off around the selected K.

4. **Design pseudocode to calculate TP, TN, FP, and FN from actual and predicted binary class labels and subsequently calculate accuracy, precision, recall, and F1-score.**

Problem Statement

Develop a procedure to evaluate a binary classification model using actual and predicted class labels. The procedure should compare the actual labels with the predicted labels and identify true positives, true negatives, false positives, and false negatives. These values are then used to calculate accuracy, precision, recall, and F1-score. The resulting metrics provide a quantitative assessment of the classification model's performance.

Algorithm / Procedure

1. Input actual and predicted binary class labels.
2. Compare each actual label with its predicted label.
3. Count TP, TN, FP, and FN.
4. Calculate accuracy.
5. Calculate precision.
6. Calculate recall.
7. Calculate F1-score.
8. Display all evaluation metrics.

Pseudocode

START

Input actual labels

Input predicted labels

Set TP = 0

Set TN = 0

Set FP = 0

```
Set FN = 0
FOR each observation
    IF actual = 1 AND predicted = 1
        TP = TP + 1
    ELSE IF actual = 0 AND predicted = 0
        TN = TN + 1
    ELSE IF actual = 0 AND predicted = 1
        FP = FP + 1
    ELSE IF actual = 1 AND predicted = 0
        FN = FN + 1
END FOR
Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2 × Precision × Recall
     / (Precision + Recall)
Display TP, TN, FP, FN
Display Accuracy, Precision, Recall, F1-score
STOP
```

Output

For example:

Confusion Matrix:

Predicted

	0	1
Actual 0	TN	FP
Actual 1	FN	TP

TP = 45

TN = 40

FP = 5

FN = 10

Accuracy = 0.85

Precision = 0.90

Recall = 0.82

F1-Score = 0.86

5. **Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.**

Problem Statement

Develop a procedure to compare different machine-learning models for a given Data Science problem. The models considered are k-Nearest Neighbors, Decision Tree, Logistic Regression, and K-Means. Appropriate evaluation measures should be selected based on whether the problem is supervised classification or unsupervised clustering. Classification models can be evaluated using accuracy, precision, recall, and F1-score, while K-Means can be evaluated using measures such as silhouette score or WCSS. The model with the most suitable performance and practical characteristics should then be selected.

Algorithm / Procedure

1. Load and preprocess the dataset.
2. Identify whether the problem is classification or clustering.
3. Split the data into training and testing sets for supervised models.
4. Train kNN, Decision Tree, and Logistic Regression.
5. Evaluate classification models using suitable metrics.
6. Apply K-Means when clustering is appropriate and evaluate using silhouette score/WCSS.
7. Compare the model performances.
8. Select the most suitable model based on performance and problem requirements.

Pseudocode

```
START
Load dataset
Preprocess dataset
IF problem is classification THEN
   Split data into training and testing sets
   Train kNN
   Train Decision Tree
   Train Logistic Regression
   Predict test data using each model
   Calculate:
```

```
        Accuracy
        Precision
        Recall
        F1-score
    Compare classification models
ELSE IF problem is clustering THEN
    Apply K-Means
    Calculate:
        Silhouette Score
        WCSS
END IF
Compare model performance
Select the most suitable model
Display evaluation results
STOP
```

Output

Example classification comparison:

Model	Accuracy	Precision	Recall	F1-Score
kNN	0.84	0.83	0.82	0.82
Decision Tree	0.88	0.87	0.89	0.88
Logistic Regression	0.86	0.85	0.84	0.84

Best Classification Model:

Decision Tree

For K-Means:

K-Means

Number of Clusters = 3

WCSS = 520

Silhouette Score = 0.61